Metroliza - Groups Analysis User Manual

Example: imagine you are comparing Machine A, Machine B, and Machine C for the same metric. This sheet helps you see whether one machine is truly running higher or lower than the others, whether the difference is large enough to matter, and whether you have enough data to trust that conclusion.

1. What this sheet is for

The Groups Analysis sheet compares the same metric across different groups and turns the results into a practical decision aid.

Question	What the sheet helps answer
Do the groups differ?	Shows whether the workbook found evidence of a difference.
How large is the difference?	Combines adjusted p-value (evidence), effect size (size), and Delta mean (direction).
Can I trust this result?	Adds caution notes and flags for small samples, imbalance, weak fit, or spec mismatch.

Best reading habit

Read the sheet as a combination of significance, effect size, direction, comparability, and caution - never from one number alone.

2. Fastest way to read the sheet

- Read the top summary to confirm scope, export level, and warnings.
- Use the Metric index to jump to the metric you care about.
- Check Metric overview first to confirm comparability and the short note.
- Check Descriptive stats to understand each group on its own.
- Check Pairwise comparisons to see exactly which groups differ.
- Use the plot, if present, as a visual cross-check rather than the only evidence.

If you only have one minute, read these first

Spec status - adj p-value - effect size - Delta mean - caution / Flags

3. What is inside each metric block

Section	What it is for
Metric overview	Short summary of the metric: spec status, shape note, and the plain-language result.
Descriptive stats	Per-group summary: sample count, average, spread, capability, fit quality, and warnings.
Pairwise comparisons	Direct group-versus-group results that show which specific groups differ from which others.

4. Column guide: Metric overview and Descriptive stats

Field	What it means
Spec status	Whether the groups can be compared under the same specification context. Read this first.
Shape note	A short comment about whether groups differ in overall data shape, not only average level.

Descriptive field	What it means
Group	The group name.
n	Number of observations in the group. Small samples usually mean lower confidence.
mean	Average value. Useful for level comparison, but sensitive to outliers.
std	Standard deviation. Shows how spread out the values are.
median	Middle value. Often more robust than the mean when data is skewed.
IQR	Spread of the middle 50% of the data.
min / max	Lowest and highest observed values.
Cp	Shows how comfortably the group's results fit inside the required specification limits.
Capability	Main capability result for the group: whether the process stays well inside the allowed limits or is too close to the edges.
Capability type	Tells you whether the capability result uses both limits or only one side of the specification.

Descriptive field	What it means
best fit model	Distribution model that best matches the observed data.
fit quality	Confidence label for that distribution fit: strong, medium, weak, or unreliable.
caution	Short warning about interpretation limits.
Flags	Compact warnings such as low sample size, imbalance, or comparability limits.

Typical guide rails for Cp / Cpk / Capability

These are common rule-of-thumb bands, not hard pass/fail laws. Always read them together with spec status, sample size, and caution notes.

Value range	Usual interpretation
Below 1.00	Not capable enough. The process is not comfortably fitting inside the limits.
1.00 to 1.33	Borderline or only just acceptable. Usually worth watching closely.
1.33 to 1.67	Generally capable / good. Often considered a comfortable working range.
Above 1.67	Strong capability. The process is staying well inside the limits.

Important: If the sheet shows a one-sided capability result, such as only one limit being used, interpret the number the same way but remember it reflects only that side of the specification.

5. Column guide: Pairwise comparisons

Field	What it means
Group A / Group B	The two groups being compared.
adj p-value	The corrected significance value - evidence that a difference is real.
effect size	How large the difference is in practical terms.
test	The statistical test used for that comparison.
Delta mean	Difference in mean between the groups. Useful because it shows direction: which group is higher or lower.
difference	Simple label telling you whether the pair is considered different.
caution	Short interpretation note for that pair.

Field	What it means
Takeaway	Plain-language conclusion.
Suggested action	Recommended next step, when provided.
Flags	Warnings relevant to the specific pair.
Why this test	Explains why this test was chosen.

Best order inside the pairwise table

adj p-value -> effect size -> Delta mean -> caution / Flags -> Takeaway / Suggested action

Typical guide rails for Adj p-value

Value range	Usual interpretation
Below 0.05	Evidence supports a real difference after correction.
0.05 to 0.10	Borderline area. Worth reading together with effect size, Delta mean, and cautions.
Above 0.10	Not enough corrected evidence for a firm difference.

6. Main result labels

DIFFERENCE Corrected evidence supports a difference.	**NO DIFFERENCE** Not enough corrected evidence for a difference.
APPROXIMATE The gap may matter, but evidence is not firm yet.	**USE CAUTION** Interpret carefully because of data quality, sample, or comparability limits.

Important: “No difference” does not automatically mean the groups are equivalent. It can also mean the data is small, noisy, or imbalanced.

7. The tests

The software selects the test automatically. Users do not need to choose it manually. It picks the safest standard option based on how the data behaves.

Test	What it checks	When it is usually chosen
Student's t-test	Compares the average of two groups.	Two groups when the data behaves normally enough and both groups have a similar spread.
Welch's t-test	Compares two averages without assuming equal spread.	Two groups when the data behaves normally enough, but the group spread is different.
Mann-Whitney U	Compares two groups using ranks.	Two groups when the standard average-based tests are not reliable enough, for example because the data is skewed, has strong outliers, or the sample is small.
ANOVA	Checks whether at least one group differs among 3+ groups.	Three or more groups when the data behaves normally enough and the groups have a similar spread.
Welch ANOVA	Global comparison for 3+ groups without assuming equal spread.	Three or more groups when the data behaves normally enough, but the group spread is different.
Kruskal-Wallis	Non-parametric global comparison for 3+ groups.	Three or more groups when the standard average-based tests are not reliable enough, for example because the data is skewed, has strong outliers, or the sample is small.
Kolmogorov-Smirnov	Checks whether two groups have a different overall distribution shape.	Two-group shape comparison.
Anderson-Darling k-sample	Checks overall distribution shape across 3+ groups.	Multi-group shape comparison.

Simple rule
If the standard tests are safe, the sheet uses them. If equal spread is doubtful, it uses the unequal-spread alternative. If the data is too skewed, too noisy, or too limited for those tests, it uses a safer rank-based test.

8. Why adj p-value and effect size must be read together

Signal	What it usually means	How to react
Low adj p-value + large effect size	Difference is statistically supported and practically meaningful.	Treat as a strong difference and review process causes.
Low adj p-value + small effect size	Difference exists, but the real-world impact may be limited.	Check whether the gap matters for decisions or capability.
No significance + noticeable effect size	There may be a real gap, but the sample may be too small or noisy.	Read caution notes and consider more data.

9. Distribution shape, spec status, and analysis level

Topic	How to use it
Distribution shape	Shows whether groups differ in overall shape, not only average. Useful when two groups have similar means but very different spread or skew.
Spec status	Read first. If specs differ across groups, capability comparison may be limited or unsafe.
Light vs Standard	Light keeps the output shorter. Standard gives fuller support, especially for pairwise and capability interpretation.

Typical spec status reading

Spec status	What it usually means
EXACT_MATCH	Best case. Groups share the same spec context and are safest to compare directly.
LIMIT_MISMATCH	Limits differ. Read pairwise and capability results more carefully.
NOM_MISMATCH	Targets differ. Do not treat the groups as fully comparable without context.

10. Safe interpretation checklist

- Are the groups fairly comparable?
- Is the sample size acceptable?
- Does the adjusted p-value support a difference?
- Is the effect size small or large?
- Does Delta mean show clear direction?
- Are there caution notes or flags?
- Do the plot and the table tell the same story?
- Does distribution shape suggest a deeper difference?
- If capability is shown, is the specification context aligned?

Avoid these shortcuts

“No significance means identical”, “Low p-value means automatically important”, and “Only capability matters” are all unsafe shortcuts.

11. Final takeaway

The Groups Analysis sheet is designed to help users make better and safer comparisons - not just faster ones.

It shows whether groups differ, how they differ, whether the difference is practically meaningful, and how much confidence should be placed in the result.

Best summary

Read the sheet as a combination of significance, effect size, direction, comparability, and caution notes.